



The Switch Task for Children: Measuring mental flexibility in young children

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Abstract

Age-related changes in mental flexibility, in the form of task switching, were assessed in 292 children (58–156 months old). Task switching was examined with a new task for young children, the Switch Task for Children (STC). The STC consists of two easy, comparable games and does not require reading skills, which makes it suitable for children younger than 6 years of age. The aims of the present study were: (1) to evaluate the STC; (2) to examine task switching in young children; and (3) to study age-related changes in task switching across childhood. The results indicated that children younger than 6 years are able to switch between two tasks and that general performance increased with age. Young children (58–89 months) displayed a larger global switch cost, that is they made more errors when the tasks were presented randomly compared with a repeated-task baseline, than older children (106–156 months). These findings suggest that the STC is a useful task for detecting task-switching abilities in young children and that the ability to maintain and manipulate two different tasks in working memory, is present, but not yet fully developed, in young children.

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1. Introduction

Executive functions encompass several inter-related processes involved in intentional, goal-directed action (Gioia, Isquith, Kenworthy, & Barton, 2002). Processes associated with executive functions are numerous, but there is a general agreement that at least the following elements are involved: anticipation, goal selection, planning, initiation of activity, self-regulation, mental flexibility, attentional control, and utilization of feedback (Anderson, 2002). These processes are

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mainly put into effect during tasks or situations that are novel, complex, or induce a conflict situation (Anderson, 1998; Zelazo, Muller, Frye, & Marcovitch, 2003).

Executive functions develop from young childhood through adolescence and are necessary for successful social and cognitive functioning. Several aspects of executive functions develop in the preschool years, such as planning (Welsh, Pennington, & Groisser, 1991), error evaluation (Kirkham, Cruess, & Diamond, 2003), attention (Espy, 1997), and goal setting (Jacques & Zelazo, 2001). However, one area that has not been thoroughly investigated in young children is the development of cognitive flexibility. Research on this topic has mainly focused on older, school-aged children (Anderson, Anderson, Northam, & Taylor, 2000; Chelune & Thompson, 1987; Jacques & Zelazo, 2001; Levin, Culhane, Hartmann, Evankovich, et al., 1991).

In order to measure cognitive flexibility, a task needs to measure the ability to shift between response sets, to learn from errors, to generate alternative strategies, to divide attention, and to concurrently process information (Anderson, 2002). A paradigm that incorporates these elements and is sensitive to developmental changes in executive functions is the task-switching paradigm (Cepeda, Kramer, & Gonzalez de Sather, 2001; Kramer, Hahn, & Gopher, 1999; Kray, Eber, & Lindenberger, 2004). This paradigm involves the performance of two relatively easy cognitive tasks (Monsell, 1996; Rogers & Monsell, 1995). In the non-switch, or 'repeated' condition, the participant is repeatedly presented one of the two tasks (e.g., AAAA or BBBB). In the other condition, the 'switch' condition, the participant switches from one task to the other (e.g., ABBAAB). Two types of task-switching costs can be derived from this paradigm. First, one can compare the mean response times (RTs) in the non-switch condition with the mean RTs in the switch condition. These switch costs may be termed global task-switching costs and are thought to reflect the difficulty of maintaining and manipulating two different mental task sets. The second method of quantifying switch costs is to examine the performance *within* the switch condition. This can be done by comparing the mean RT for trials in which a switch was demanded (e.g., ABBAAB) with the mean RT for trials that do not require a switch (e.g., ABBAAB). These so-called local switch costs are an indication of the executive functions associated with the actual task switching (Verhaeghen & Basak, 2005; Verhaeghen & Cerella, 2002).

Several researchers have shown that the global switch costs are more sensitive to age differences than local costs (Kray et al., 2004; Kray & Lindenberger, 2000; Verhaeghen & Cerella, 2002). This might be due to the difference in control processes used. Global switching requires the simultaneous maintenance of two mental task sets versus one mental set; local switching involves the selection between two sets that are already loaded in working memory. This interpretation suggests that the ability to efficiently maintain and manipulate two tasks sets in working memory instead of one is more affected by age than the ability to perform the task switch itself (Kray & Lindenberger, 2000; Verhaeghen & Basak, 2005).

The influence of age on task switching has mainly been examined in adults. Only a few studies have examined age-related switch costs in children (Cepeda et al., 2001; Kramer, Cepeda, & Cepeda, 2001; Kray et al., 2004). Kray et al. examined age-related changes in task switching in children (mean: 9.4 years), young adults (21.5 years), and older adults (65.3 years). Cepeda et al. presented a switching paradigm to individuals ranging in age from 7 to 82 years. In the task of Cepeda et al., the participants were presented with a row of digits and were asked to either indicate the number or the value of the digits. In the experiment of Kray et al., the participants had to categorize pictures by object (animal–fruit) or by color (gray-colored). The task that had to be performed, object or color, was indicated by semantic instructional cues. Unfortunately, the above-mentioned tasks are not suitable for young children because the tasks require a fluent reading response, which is not developed in children younger than 6 years of age.

The present study focuses on task switching in children younger than 6 years and the development of this ability across childhood. The Cognitive Complexity and Control theory (Zelazo & Frye, 1998) hypothesizes that as children grow older, they are able to formulate and use an increasingly complex rule structure. At the age of 4–5 years, children represent a rule structure that comprises higher-order rules, for example conditional ‘if–then rules’, and should, therefore, be able to perform a switching task. In order to assess the ability of this age group to switch between two tasks, we developed a new task, the Switch Task for Children (STC). To appeal to young children the STC is presented as a game. Two easy, yet conflicting, tasks are presented, inducing a conflict situation, which is known to draw on mental flexibility processes (Zelazo et al., 2003). The tasks do not require a reading response and both local and global task-switching costs can be derived.

To summarize the specific goals of the present study: First, to develop and evaluate a new switching task, the STC. Second, to examine whether children younger than 6 years indeed possess the executive functions necessary to perform a task-switching paradigm. Our third aim is to study age-related changes in task switching across childhood.

2. Method

2.1. Participants

Two hundred ninety-two healthy children were recruited from infant and primary schools in the Maastricht area in the Netherlands. Permission for recruitment was gained from the head teacher of the schools, and written informed consent for the participation of the children was obtained from their parents. In total the data for 232 (96 males; 136 females) children were included in the analyses (described later). The main reasons for exclusion were: developmental or learning problems ($n = 18$), motor problems ($n = 15$), and interruptions or technical problems ($n = 12$). There was no sex difference among the excluded participants (20% of the females; 21% of the males). Only two girls aged 61 and 67 months in the youngest group were unable to learn the task adequately. The youngest girl did not finish the switch training, and; the oldest girl was not able to learn the second task (see below). The remaining participants were divided into five age categories (58–156 months) (see Table 1).

All children received a small present for participation. The youngest group received a certificate picturing a prince, princess, and a treasure chest, and the older children were given a plastic cube containing an “ability game.” Additional information about each participant (e.g., developmental

Table 1
Descriptive data of the participants

Age group	Age range (months)	Age in months		Males	Females	Total participants
		<i>M</i>	<i>S.D.</i>			
Group 1	58–73	67.45	3.94	22	25	47
Group 2	74–89	80.33	4.66	22	30	52
Group 3	90–105	96.07	3.63	22	21	43
Group 4	106–121	115.11	4.31	13	25	38
Group 5	>122	138.42	10.19	17	35	52
Total	58–156	99.35	26.67	96	136	232

problems, learning problems, etc.) was obtained by sending the parents a questionnaire after all participants had been tested.

2.2. *Materials*

The Switch Task for Children, see below for a detailed description, was run on a laptop computer attached to a 17-in. color monitor. The software used was programmed in E-prime (Psychology Software Tools, <http://www.pstnet.com/>). Two buttons, one for the left hand and one for the right hand, recorded the responses of the participants.

2.3. *Procedure*

The STC is a computer game and consists of an instruction phase, a training phase, and a test phase. The order of task presentation during training and the correctness of response are counterbalanced, leading to four different task versions. For the sake of clarity, only one version of the task is described here.

2.3.1. *Instruction phase*

On entering the room, the child was asked to sit in front of the computer monitor and to hold the response buttons in a comfortable position. The experimenter started a recorded fairy tale containing task information. While listening to the story, the participant viewed several scenes of the tale on the monitor. The ‘story’ was, that a prince and princess had been robbed. The participant was asked to help them to find their stolen treasure chests.

Three practice trials were presented to familiarize the participant with the general task procedure. In these trials an orange house and a blue house with a white cat in between them were presented against a daytime sky. The left and right position of the two houses varied over trials. The treasure was hidden in one of these houses, and the participant was encouraged to select one of the two houses by pressing the accompanying button. No time limit was set for responding. The correctness of the participant’s choice was presented on a feedback screen (1500 ms). Selection of the orange house, the correct response, was ‘rewarded’ by replacing the house with a treasure chest. Selection of the blue house, the incorrect response, was ‘punished’ by marking this house with a large pink cross. After presentation of the feedback screen, there was an inter-trial interval during which a white cat was displayed on a green lawn (1000 ms). During the training and practice trials, the participant was verbally encouraged (e.g., ‘you are doing very well’) and, if necessary, clues to solve the task were given (e.g., the treasure chest is always in the same house). No clues were given on test trials and verbal feedback was minimized.

2.3.2. *Training Task 1*

The task presented in Phase 1 was similar to the one presented during the three practice trials. A time limit of 4000 ms per trial was set for the remainder of the experiment. If the participant failed to respond within this time limit, the trial was considered incorrect, and no feedback was provided. The duration of the inter-trial interval depended on the response latency of the participant. The duration was calculated by adding the difference between the time limit and the response latency to the inter-trial interval of 1000 ms. If the participant did not respond, the inter-trial interval lasted 2500 ms. In this way, each trial had a fixed duration of 6500 ms (i.e., response latency + feedback screen + inter-trial interval).

The left and right positions of the two houses were mixed pseudo-randomly, with the restriction that the position was not the same on more than two trials in a row. The next phase, Phase 2, started as soon as the participant fulfilled one of the following criteria: seven correct trials in a row, no more than 1 incorrect trial out of 10 trials, or no more than 2 incorrect trials out of 13 trials ($p < .01$ for reaching one of these criteria by chance). The experiment was discontinued if the participant did not reach one of these criteria within 60 trials.

2.3.3. Training Task 2

Task 2 strongly resembled Task 1 with the following exceptions. Instead of presenting the houses against a daytime sky, they were presented against a night-time sky. Furthermore, selection of the blue house instead of the orange house was rewarded; selection of the orange house led to the cross sign. All further details, such as stimulus duration, criteria, and restrictions, were the same as mentioned in Task 1. Reaching one of the criteria automatically led to the switch training.

2.3.4. Switch training

The two experimental conditions (switch and non-switch) were introduced during the Switch Training. In the non-switch condition, one task was repeatedly presented, either Task 1 (day) or Task 2 (night). In the switch condition, trials of Task 1 and Task 2 were randomly mixed and presented in a ratio of 1 to 1. The experimental conditions were presented in blocks of four trials. Each block was preceded by one of three signal screens presented for 3000 ms that informed the participant about the upcoming task (day, night, or a mix of day and night). In the first two blocks, a non-switch condition was presented (day and night, respectively), but in the third block the switch condition was introduced. Thereafter, blocks of day and night were interspersed with blocks containing the switch condition. The criteria did not come in effect until block 4 of the switch training. In this way, the participant had viewed each of the three signal screens before the Test Phase started. All other details were the same as above. After one of the criteria was reached, a picture of the robber popped up and children could take a short break, if necessary.

2.3.5. Test phase

The test consisted of eight blocks of 10 trials. The order of presentation of the tasks was identical for each participant: day–switch–night–switch–day–switch–night–switch. The corresponding signal screen preceded each of these test blocks. The test started with a non-switch condition. The reason for this was to avoid a mixture of task-switching costs and a warm-up decrement due to the interruption of the task by the picture of the robber. During the Test Phase, the white cat on the screen changed color to keep the participant motivated and to give an indication of the remaining test time. The experiment ended with a picture of the prince, the princess, and their treasure chests.

2.4. Statistics

Only the data of the Test Phase were used for data analyses. Separate data analyses were carried out to focus on the performance of the youngest group children (Group 1). This group was of particular interest because no data in the task-switching literature of this group is available. For each test trial, both the time to press a button (response time, RT, in ms) and the correctness of the button press (correct/incorrect) were registered.

Only RTs of correct responses were used for data analyses. Responses faster than 200 ms (false alarms) were considered incorrect and excluded from data analyses. The percentage of false alarms was very low (0.06%) and was not further analyzed. The number of errors is expressed as

a percentage and the RT as an average. The data and accompanying analyses were subdivided into global and local data. Global data concern the RTs and percentages of errors in the switch and non-switch conditions. The local data concern the trials within one condition, the switch condition.

2.4.1. Switch costs RTs

Switch costs were determined as global RT switch costs ($\text{GlobCost}_{\text{RT}}$) and local RT switch costs ($\text{LocCost}_{\text{RT}}$). The former was calculated by subtracting the mean RTs in the non-switch condition from the mean RTs in the switch condition. The latter was calculated by subtracting the mean RTs of the non-switch trials (e.g., ABBAAB) in the switch condition from the mean RTs of the switch trials (e.g., ABBAAB) in the switch condition (Wainer, 1991). The advantage of using the latter method is that effects of arousal or other between-block factors are eliminated (Cepeda et al., 2001; Verhaeghen & Cerella, 2002); however, the first method is more sensitive in detecting age-differences (Kray et al., 2004; Verhaeghen & Cerella, 2002).

Possible baseline differences among the age groups were adjusted by subtracting the baseline performance (non-switch condition or non-switch trials) from the switch performance (switch condition or switch trials). This subtraction method assumes that, for all age groups, if the switch performance does not require additional executive control processes, the switch performance will be equal to the non-switch performance. Because in the present study each switch performance is preceded by an accompanying non-switch performance, canceling out factors as fatigue or inattention, this seems to be plausible assumption and therefore, this correction method was used (see for a comparison of other adjustment methods also Wainer, 1991).

2.4.2. Switch costs errors

Global error switch costs are expressed as the difference in errors between the non-switch and switch conditions ($\text{GlobCost}_{\text{err}}$). Local error switch costs were calculated by subtracting the percentage of errors the non-switch trials in the switch condition from the percentage errors on the switch trials in the switch condition ($\text{LocCost}_{\text{err}}$).

Data were analyzed using parametric tests (analysis of variance [ANOVA], and GLM with repeated measures). In the GLMs repeated measures analysis, experimental condition or trial type functioned as within-subject factors and age group functioned as between-subject factor. Bonferroni tests were used to adjust the significance level when multiple or pair wise comparisons were made. Pearson correlation coefficients were calculated between age (expressed in months) and RTs or percentage of errors. The rejection criterion was $p < .05$.

3. Results

Separate GLMs repeated measures analyses were performed to examine the possible effect of sex and test version on the RTs and error data. In each analysis, experimental condition or trial type (switch and non-switch) functioned as within-subject factors and sex or test version functioned as between-subject factors. These analyses did not reveal a global or local effect of test version, $F_s(3, 228) < 1.05$, or sex, $F_s(1, 230) < 2.97$ (Fig. 1).

3.1. Response time (RT) analyses

3.1.1. Group 1: RT analyses

The data of Group 1 are presented on the left side of each graph. Two separate GLMs repeated measures analyses were performed on the global (upper left panel) and local RT data (upper right

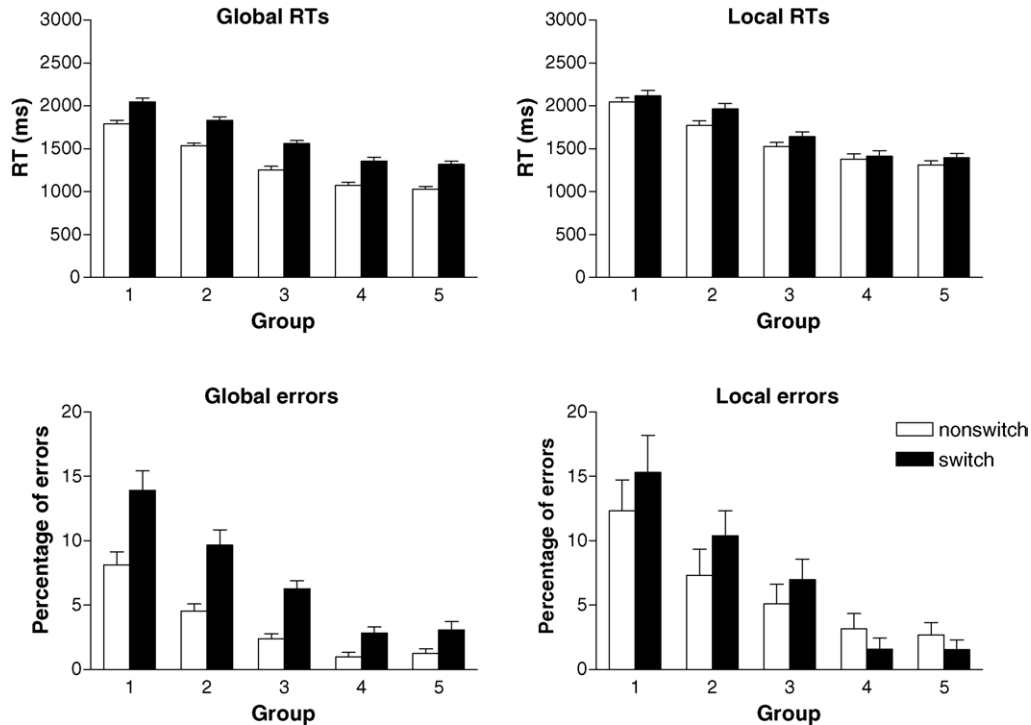


Fig. 1. Global (left side) and local switch costs (right side) as a function of age group. RT costs (ms) are displayed in the top panels; error costs (%) are displayed at the bottom. Error bars refer to the standard errors.

panel). Both the analysis of the global RTs and the analysis of the local RTs revealed a main effect of experimental condition, $F(1, 46) = 9.76, p < .005$. The children responded slower in the switch than in the non-switch condition, with mean RTs of 2044 and 1788 ms, respectively. Similarly, they responded slower on the switch trials (2275 ms) than on the non-switch trials (2068 ms) in the switch condition.

3.1.2. Age differences: RT analyses

3.1.2.1. Global RTs. A GLM repeated measures analysis revealed a main effect of experimental condition, $F(1, 227) = 802.48, p < .001$, and a main effect of age group, $F(4, 227) = 73.40, p < .001$. As expected, children were slower in the switch than in the non-switch condition. Furthermore, the RTs of the children in Groups 1, 2, and 3 differed from each other and from the older groups, $p < .01$, with the younger children being slower than the older children. No difference was found between the RTs of the two oldest groups, Group 4 and Group 5, $p = 1.00$. Unexpectedly, the interaction between age group and experimental condition found in similar developmental studies was not observed, $F(4, 227) < 1$.

3.1.2.2. Local RTs. Analysis of the local RTs also revealed a main effect of trial type, $F(1, 227) = 20.23, p < .001$, and a main effect of age, $F(4, 227) = 42.54, p < .001$. As for the global RTs, children responded slower on the switch than the non-switch trials. Similarly, the RTs of the children in Groups 1, 2, and 3 differed from each other and from the older groups, $p < .05$.

The RTs of the two oldest age groups were not significantly different, $p = 1.00$. No interaction between age group and trial type was observed, $F(4, 227) = 1.44$.

The lack of an interaction between experimental condition/trial type and age group in both the global and local analyses indicated that the RT switching costs did not change with increasing age. This was confirmed by the absence of a correlation between age, expressed in months, and $\text{GlobCost}_{\text{rt}}$ or $\text{LocCost}_{\text{rt}}$, $r = .025$ and $-.048$, respectively.

3.2. Error rate analyses

3.2.1. Group 1: error rate analyses

Two separate GLMs repeated measures analyses were performed on the global (lower left panel) and local error data (lower right panel). The global analyses revealed an effect of experimental condition, $F(1, 46) = 26.42$, $p < .001$. The proportion of errors was 13.9% in the switch condition and 8.1% in the non-switch condition. Although more local errors were made in the switch trials (15.3% versus 12.3%), the difference was not significant, $F(1, 46) = 1.41$.

3.2.2. Age differences: error rate analyses

3.2.2.1. Global errors. A GLM repeated measures on the global error data revealed a main effect of experimental condition, $F(1, 227) = 92.72$, $p < .001$, with more errors being made during the switch (7.3%) than during the non-switch (3.5%) condition. Furthermore, there was a main effect of age group, $F(4, 227) = 28.39$, $p < .001$, and a significant interaction between experimental condition and age group was found, $F(4, 227) = 4.64$, $p < .005$. The effect of age group was analyzed further and revealed that the two youngest groups, Group 1 and Group 2, significantly differed from each other and the older groups, Groups 4 and 5, $ps < .05$. No other differences between the age groups were found. The interaction found was caused by a differential effect of experimental condition on the age groups. A separate GLM univariate analysis on the difference score between the switch and non-switch conditions, $\text{GlobCost}_{\text{err}}$, confirmed this observation, $F(4, 227) = 4.64$, $p < .005$. Post hoc comparisons revealed that the $\text{GlobCost}_{\text{err}}$ in the two youngest groups was larger than the $\text{GlobCost}_{\text{err}}$ in the two oldest groups, $ps < .05$. No evidence for a speed-accuracy trade-off was found. Error rates covaried such that the error rate was lower with faster responses.

3.2.2.2. Local errors. The local error data only revealed a main effect of age group, $F(4, 227) = 10.70$, $p < .001$. Post hoc analyses revealed that the children in Group 1 made more errors than the children in the Groups 3, 4, and 5, $ps < .005$. The children in Group 2 made more errors than the children in Groups 4 and 5, $ps < .05$. No age group $\times$ trial type interaction was found, $F(4, 227) = 1.32$.

A separate analysis revealed a negative correlation between age (expressed in months) and $\text{GlobCost}_{\text{err}}$, $r = -.26$, $p < .001$. The correlation analyses between age and $\text{LocCost}_{\text{err}}$ pointed in the same direction, but was not significant, $r = -.12$, $p = .078$. These data indicate that the difference in errors made in the switch and non-switch conditions decreases with increasing age.

4. Discussion

This study was performed to examine the ability of children younger than 6 years of age to switch between two tasks, and to examine the development of this ability across primary school ages. To this end we developed a new task, the Switch Task for Children. The STC is especially

suitable for young children because the task is presented in the form of an adventurous game and reading skills are not needed. In the STC, children learned to select one of two houses during the first task, and to select the alternative house in the second task. The two tasks were distinguishable by two different sky scenes, day and night, against which the houses were presented. Subsequently, either the same task was presented repeatedly, non-switch condition, or the two tasks were randomly mixed, switch condition.

The first important result to note is that more than 96% of the children younger than 6 years and 100% of the children older than 6 years, were able to master the task. Thus, the STC is suitable for children as young as 58 months, but can also be administered to older children. Furthermore, the STC can detect both global and local switch costs. The global switch costs can be derived by comparing the switch and non-switch conditions; the local data provide more detailed information about the switch and non-switch trials in the switch condition. Global switch costs are thought to reflect mental processes that are involved in maintaining and manipulating two mental task sets, the executive processes associated with actual task switching are expressed in the local switch costs (Verhaeghen & Basak, 2005).

Second, our data are very similar to those for other more traditional switching tasks in older children and adults (Allport, Styles, & Hsieh, 1994; Kray et al., 2004; Kray & Lindenberger, 2000). That is, more errors and slower responses were made in the switch condition than in the non-switch condition. These global switch costs indicate that it is more difficult to maintain and manipulate two mental task sets (switch condition) than one task set (non-switch condition). The local data, associated with the actual task switching, were in part consistent with our global data, in that responses were slower in the switch trials than in the non-switch trials, however, the type of trial did not affect the proportion of incorrect responses.

The data for the youngest group (58–73 months) are particularly interesting. As mentioned above, more than 96% of the children younger than 6 years of age were able to perform the STC. They showed a pattern of responses similar to that of older children and adults, namely, they made more errors and responded slower during the switch condition than during the non-switch condition. Thus, even children younger than 6 years of age have access to executive functions that enable them to switch between two conflicting tasks in a similar way as older children and adults do. These results are in accordance with the Cognitive Complexity and Control theory (Zelazo & Frye, 1998), which predicts that children aged about 4–5 years possess a rule structure that contains higher-order ‘if-then rules.’ However, the possession of such a rule structure in young children has, to our knowledge, not been tested in a true switching paradigm. Therefore, the present data add to the existing literature on the development of executive functions, more specifically mental flexibility, in children younger than 6 years.

Developmental differences were observed in the global and local data. With increasing age, the children made fewer errors and responded faster. The performance of the youngest groups, Groups 1–3, deviated from each other and from that of the older groups, Groups 4 and 5. No developmental differences were observed between the two oldest groups the differences found mainly reflect an overall improvement in cognitive and/or motor skills. As children grow older, cognitive skills emerge and become increasingly refined and elaborated (Sroufe, Cooper, DeHart, Marshall, & Bronfenbrenner, 1992). This will make the STC easier for older children than for younger children. Similarly, the more developed motor skills will help the older children to make faster and more accurate responses. Another developmental difference was found in the switch costs. This difference was only observed in the global data, but not in local data, and is in line with previous studies on age-related changes in task switching (Kray et al., 2004; Kray & Lindenberger, 2000; Verhaeghen & Cerella, 2002). However, unlike the abovementioned studies, the difference

was only found in the global error data, $\text{GlobCost}_{\text{err}}$; no switch cost differences were found for the global RT data, $\text{GlobCost}_{\text{rt}}$. More specifically, children in the two youngest groups, Groups 1 and 2, made relatively more errors in the switch condition than in the non-switch condition, than did children in the older groups, Groups 4 and 5. Separate correlation analyses confirmed a negative relation between $\text{GlobCost}_{\text{err}}$ and age in months, thus the difference between the switch and non-switch conditions decreased with age. These results suggest that the executive functions that are necessary to maintain and manipulate two tasks sets in working memory are present, but not yet fully developed by the age of 89 months.

The observed lack of an age difference in global RT costs is unexpected and can have several causes. One possible explanation lies in the task used. Other studies using a switch paradigm require a reading or counting response (Cepeda et al., 2001; Kramer et al., 1999; Kray et al., 2004). One could argue that the task is much harder for younger children than for older children. Although the results are not always consistent, some authors do find greater RT switch costs when task complexity increases (Rubinstein, Meyer, & Evans, 2001). The tasks used in the present study were relatively easy considering the low percentage of errors in each age group. This might explain the absence of an age effect on the Cost_{rt} . Another explanation concerns how omitted responses (RTs longer than 4000 ms) were handled. These responses were considered incorrect and excluded from the analyses. Replacing these omitted responses by the maximum RT value of 4000 revealed a main effect of age on the local RT costs, $F(4, 227) = 2.51, p < .05$. Larger switch costs were observed in Group 2 (mean: 260 ms) compared to Group 4 (mean: 20 ms), $p < .05$. No age differences were detected in the $\text{GlobCost}_{\text{rt}}$, $F(4, 227) < 1$. The reason for the absence of an age effect in the global RT costs is unclear, especially because global RT switch costs are the most sensitive measure to detect age differences in switch costs (Kray et al., 2004; Verhaeghen & Cerella, 2002).

Our data are in the main consistent with those of both Cepeda et al. (2001) and Kray et al. (2004). These authors investigated whether children aged 7 years and older were able to correctly perform a switch task and expressed switch costs in the form of an increase in errors and RTs. Additionally, the present study includes a group of children younger than 6 years. Because the STC is suitable for children without a reading response, a next step would be to test even younger, preschool, children, for example aged 3–4 years. This would make it possible to determine when children gain the ability to switch between two tasks.

The study has some limitations, the main one being the rather narrow age range studied. Expanding the age range by testing young adults and a population of older adults might help to reconstruct the U-shaped function between age and switch costs observed in other studies (Cepeda et al., 2001; Kray et al., 2004). In a future study it would be interesting not only to expand the age range, but also to administer the STC to children with executive function problems, for example attention deficit/hyperactivity disorder (Cepeda, Cepeda, & Kramer, 2000; Kramer et al., 2001), or to vary the task complexity to further explore the relation between switch costs, age, and task complexity (Verhaeghen & Cerella, 2002). Finally, the STC is designed in such way that performance of the task in a neuroimaging environment is possible. Each trial has the same total duration, no verbal response is required, and test duration is minimized. It would be interesting to relate the maturation of the prefrontal cortex, which is known to be involved in executive functions, with task performance (Casey, Giedd, & Thomas, 2000; Dove, Pollmann, Schubert, Wiggins, & von Cramon, 2000; Durston et al., 2002).

To summarize, the new STC appears to be a useful task to detect switching abilities in young children. The results indicate that children younger than 6 years of age possess the cognitive skills to switch between two relatively easy tasks. The results are generally in line with previous

task-switching experiments. On average, young children made more errors, responded slower, and had larger switch costs than older children. Future directions for the present study include presentation of the STC in a neuroimaging environment, administration of the task to even younger, preschool children, and presentation of the STC to children with executive function problems.

References

- Allport, D. A., Styles, E. A., & Hsieh, S. (1994). Shifting intentional set: Exploring the dynamic control of tasks. In C. Umiltà & M. Moscovitch (Eds.), *Attention and performance 15: Conscious and nonconscious information processing. Attention and performance series* (pp. 421–452). Cambridge, MA, USA: The MIT Press.
- Anderson, P. (2002). Assessment and development of executive function (EF) during childhood. *Neuropsychology, Development, and Cognition. Section C, Child Neuropsychology*, 8(2), 71–82.
- Anderson, P., Anderson, V., Northam, E., & Taylor, H. G. (2000). Standardization of the contingency naming test (CNT) for school-aged children: A measure of reactive flexibility. *Clinical Neuropsychological Assessment*, 1, 247–273.
- Anderson, V. (1998). Assessing executive functions in children: Biological, psychological, and developmental considerations. *Neuropsychological Rehabilitation*, 8(3), 319–349.
- Casey, B. J., Giedd, J. N., & Thomas, K. M. (2000). Structural and functional brain development and its relation to cognitive development. *Biological Psychology*, 54(1–3), 241–257.
- Cepeda, N. J., Cepeda, M. L., & Kramer, A. F. (2000). Task switching and attention deficit hyperactivity disorder. *Journal of Abnormal Child Psychology*, 28(3), 213–226.
- Cepeda, N. J., Kramer, A. F., & Gonzalez de Sather, J. C. (2001). Changes in executive control across the life span: Examination of task-switching performance. *Developmental Psychology*, 37(5), 715–730.
- Chelune, G. J., & Thompson, L. L. (1987). Evaluation of the general sensitivity of the Wisconsin Card Sorting Test among younger and older children. *Developmental Neuropsychology*, 3(1), 81–89.
- Dove, A., Pollmann, S., Schubert, T., Wiggins, C. J., & von Cramon, D. Y. (2000). Prefrontal cortex activation in task switching: An event-related fMRI study. *Brain Research. Cognitive Brain Research*, 9(1), 103–109.
- Durston, S., Thomas, K. M., Yang, Y., Ulug, A. M., Zimmerman, R. D., & Casey, B. J. (2002). A neural basis for the development of inhibitory control. *Developmental Science*, 5(4), F9–F16.
- Espy, K. A. (1997). The Shape School: Assessing executive function in preschool children. *Developmental Neuropsychology*, 13(4), 495–499.
- Gioia, G. A., Isquith, P. K., Kenworthy, L., & Barton, R. M. (2002). Profiles of everyday executive function in acquired and developmental disorders. *Neuropsychology, Development, and Cognition. Section C, Child Neuropsychology*, 8(2), 121–137.
- Jacques, S., & Zelazo, P. D. (2001). The Flexible Item Selection Task (FIST): A measure of executive function in preschoolers. *Developmental Neuropsychology*, 20(3), 573–591.
- Kirkham, N. Z., Cruess, L., & Diamond, A. (2003). Helping children apply their knowledge to their behavior on a dimension-switching task. *Developmental Science*, 6(5), 449–476.
- Kramer, A. F., Cepeda, N. J., & Cepeda, M. L. (2001). Methylphenidate effects on task-switching performance in attention-deficit/hyperactivity disorder. *Journal of the American Academy of Child and Adolescent Psychiatry*, 40(11), 1277–1284.
- Kramer, A. F., Hahn, S., & Gopher, D. (1999). Task coordination and aging: Explorations of executive control processes in the task switching paradigm. *Acta Psychologica*, 101(2/3), 339–378.
- Kray, J., Eber, J., & Lindenberger, U. (2004). Age differences in executive functioning across the lifespan: The role of verbalization in task preparation. *Acta Psychologica*, 115(2/3), 143–165.
- Kray, J., & Lindenberger, U. (2000). Adult age differences in task switching. *Psychological Aging*, 15(1), 126–147.
- Levin, H. S., Culhane, K. A., Hartmann, J., Evankovich, K., Mattson, A. J., Harward, H., et al. (1991). Developmental changes in performance on tests of purported frontal lobe functioning. *Developmental Neuropsychology*, 7(3), 377–395.
- Monsell, S. (1996). Control of mental processes. In V. Bruce (Ed.), *Unsolved mysteries of the mind: Tutorial essays in cognition* (pp. 93–148). Oxford, UK: Erlbaum (UK) Taylor & Francis.
- Rogers, R. D., & Monsell, S. (1995). Costs of a predictable switch between simple cognitive tasks. *Journal of Experimental Psychology: General*, 124(2), 207–231.
- Rubinstein, J. S., Meyer, D. E., & Evans, J. E. (2001). Executive control of cognitive processes in task switching. *Journal of Experimental Psychology: Human Perception and Performance*, 27(4), 763–797.

- Sroufe, L. A., Cooper, R. G., DeHart, G. B., Marshall, M. E., & Bronfenbrenner, U. (1992). *Child development: Its nature and course* (2nd ed.). New York, NY, England: McGraw-Hill Book Company.
- Verhaeghen, P., & Basak, C. (2005). Ageing and switching of the focus of attention in working memory: Results from a modified N-Back task. *Quarterly Journal of Experimental Psychology: Human Experimental Psychology*, 58A(1), 134–154.
- Verhaeghen, P., & Cerella, J. (2002). Aging, executive control, and attention: A review of meta-analyses. *Neuroscience and Biobehavioral Reviews*, 26(7), 849–857.
- Wainer, H. (1991). Adjusting for differential base rates: Lord's paradox again. *Psychological Bulletin*, 109(1), 147–151.
- Welsh, M. C., Pennington, B. F., & Groisser, D. B. (1991). A normative-developmental study of executive function: A window on prefrontal function in children. *Developmental Neuropsychology*, 7(2), 131–149.
- Zelazo, P. D., & Frye, D. (1998). Cognitive complexity and control: II. The development of executive function in childhood. *Current Directions in Psychological Science*, 7(4), 121–126.
- Zelazo, P. D., Muller, U., Frye, D., & Marcovitch, S. (2003). The development of executive function: Cognitive complexity and control—Revised. *Monographs of the Society for Research in Child Development*, 68(3), 93–119.